

Mod 3 & 4

Books:

Manufacturing Science
by
Chosh & Mallik

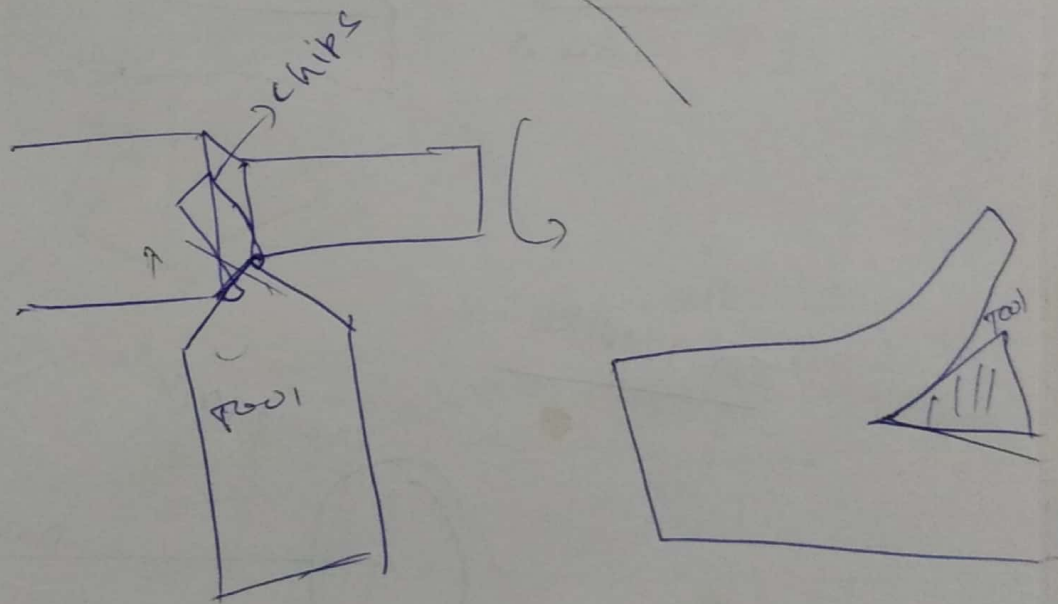
Casting
Forming
Joining
Machining.

Rao

Cutting tools, Machine tool
↓ ↓
drill bit drilling machine.

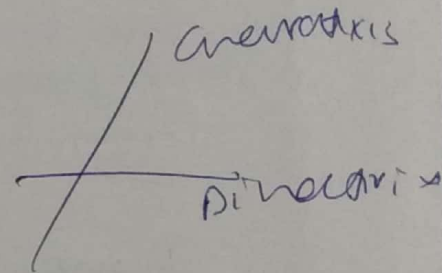
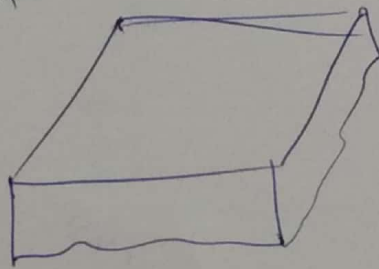
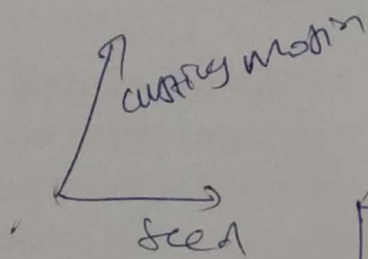
Turning

lathe

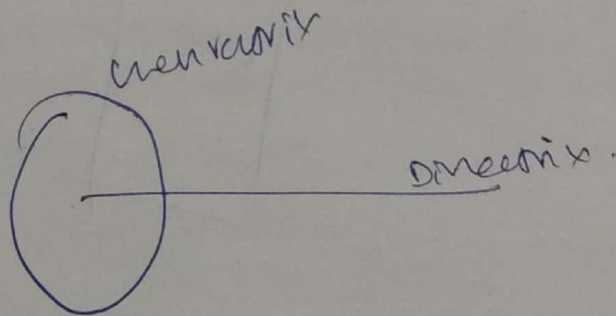


Depending on the nature of two relative motions, various types of surfaces can be produced. To explain this in more detail, consider a point P where the material is being cut at a particular distance. Now for the cutting motion

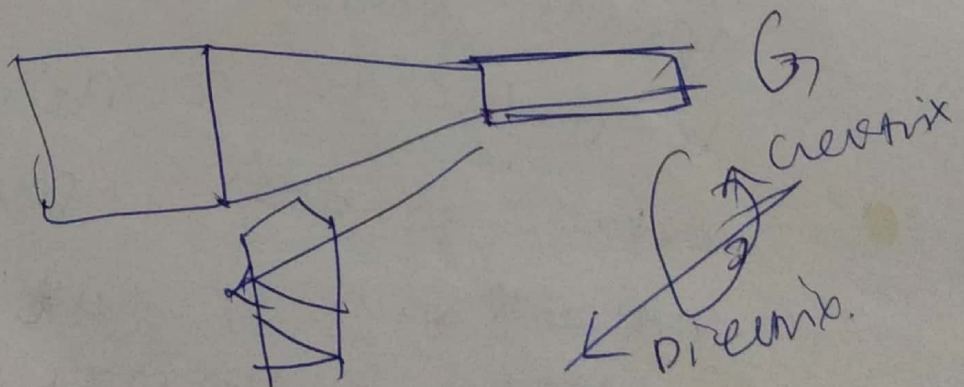
is rectilinear and the feed is also rectilinear. the machine surface will be plane. The line generated by the cutting motion is called generatrix, the line from the feed motion is called directrix.

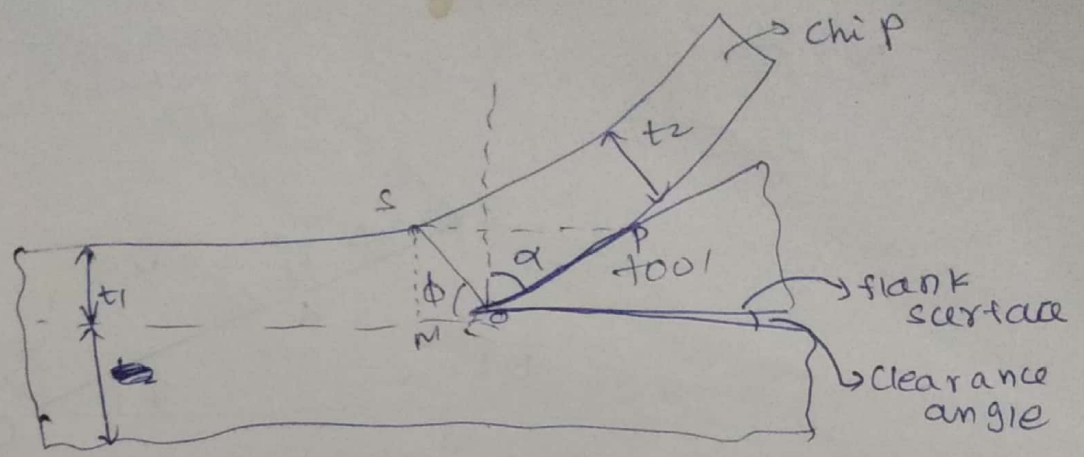


In cone



Taper turnings

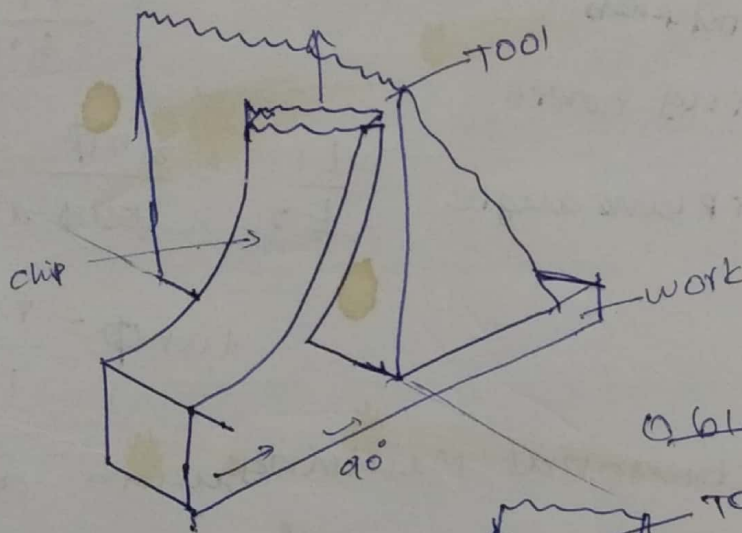




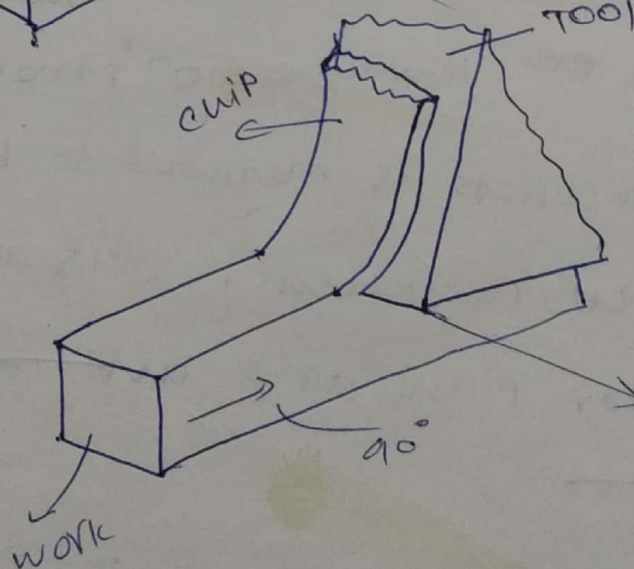
t_1 = uncut thickness
 t_2 = chip thickness

so - shear plane
 α - rake angle

orthogonal machining



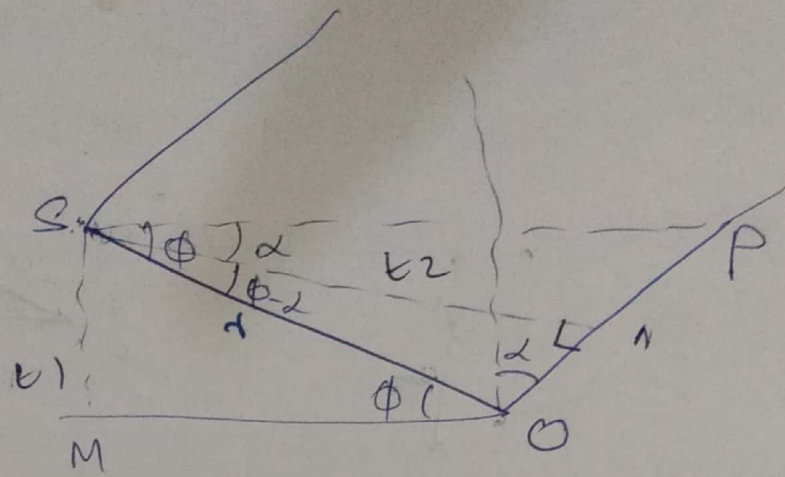
Oblique cutting
 or
Oblique Machining
 (almost all)



For purpose of analysis, orthogonal machining.

Refer
 Sen &
 Bhattacharya

Analysis



$$OS = \frac{SV}{\cos(\phi - \alpha)} = \frac{t_2}{\cos(\phi - \alpha)}$$

t_1 - uncut thickness
 t_2 - chip thickness

r - cutting ratio

ϕ - shear plane angle

$$= \frac{SV}{\sin \phi} = \frac{t_1}{\sin \phi}$$

$$\frac{t_1}{t_2} = \frac{\sin \phi}{\cos(\phi - \alpha)} = r$$

$$\tan \phi = \frac{r \cos \alpha}{1 - r \sin \alpha}$$

Q)

During orthogonal machining, with a cutting tool having a 10° rake angle, the chip thickness is measured to be 0.4 mm and the uncut thickness is 0.15 mm. Determine the shear plane angle and ~~also the magnitude of shear~~

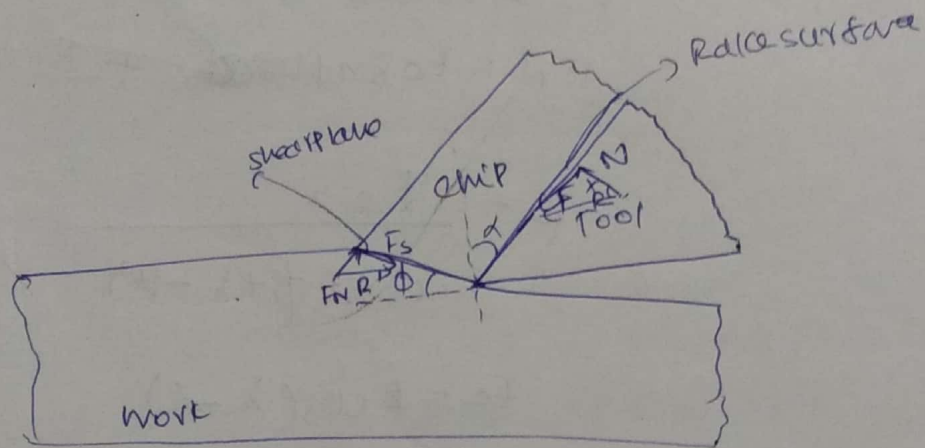
$$\phi = \tan^{-1} \left(\frac{r \cos \alpha}{1 - r \sin \alpha} \right)$$

$$r = \frac{t_1}{t_2} = \frac{0.15}{0.4} = \frac{3}{8} = 0.375$$

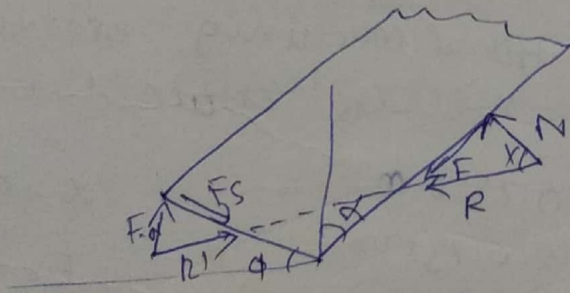
$$= \tan^{-1} \left(\frac{0.37 \cos 10^\circ}{1 - 0.37 \sin 10^\circ} \right)$$

$$\phi = 21.27^\circ$$

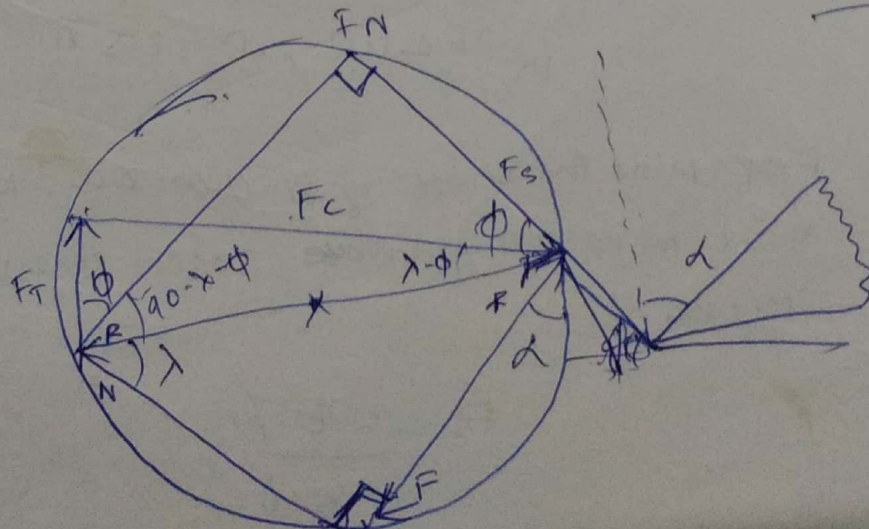
Mechanics of chip formation: (Emst & Merchant)



we will consider chip as a rigid body to analyse.



Merchant's circle
Theorem



$$F_c = F_s \cos \phi + F_N \sin \phi$$

$$F_T = F_N \cos \phi + F_s \sin \phi$$

$$F = F_c \sin \alpha + F_T \cos \alpha$$

$$N = F_c \cos \alpha - F_T \sin \alpha$$

$$F_s = F_c \cos \phi - F_T \sin \phi$$

$$F_N = F_c \sin \phi + F_T \cos \phi$$

$$R = \frac{F_s}{\cos(\phi + \lambda - \alpha)}$$

$$F_c = R \cos(\lambda - \alpha)$$

$$F_T = R \sin(\lambda - \alpha)$$

$$F_s = \frac{W_t Z}{\sin \phi}$$

9) During an orthogonal machining operation on mild steel, the results obtained are

$$t_1 = 0.25 \text{ mm}$$

$$\alpha = 0^\circ$$

$$t_2 = 0.75 \text{ mm}$$

$$F_c = 950 \text{ N}$$

$$\text{width of cut, } w = 2.5 \text{ m}$$

$$F_T = 475 \text{ N}$$

Determine the coefficient of friction b/w tool and chip.
Determine the ultimate shear stress of the work material.

$$F_s = \frac{W_t Z}{\sin \phi}$$

$$F_s = F_L \cos \alpha - F_T \sin \alpha$$

$$F_s = 950 \text{ or } (-475 \text{ N})$$

$$F_s = 950$$

$$\mu = \frac{F}{N}$$

- Q) Mild steel is being machined at a cutting speed of 200 m/min with a tool of rake angle 20° . The width of cut & uncut thickness are 2 mm & 0.2 mm resp. If the avg. value of the coeff of friction b/w tool & chip is 0.5 & shear stress τ_s of work material is 400 N/mm². Determine the shear angle and the cutting & thrust components of MC.

$$2\phi + \lambda = 90^\circ$$

$$\tan \lambda = \mu$$

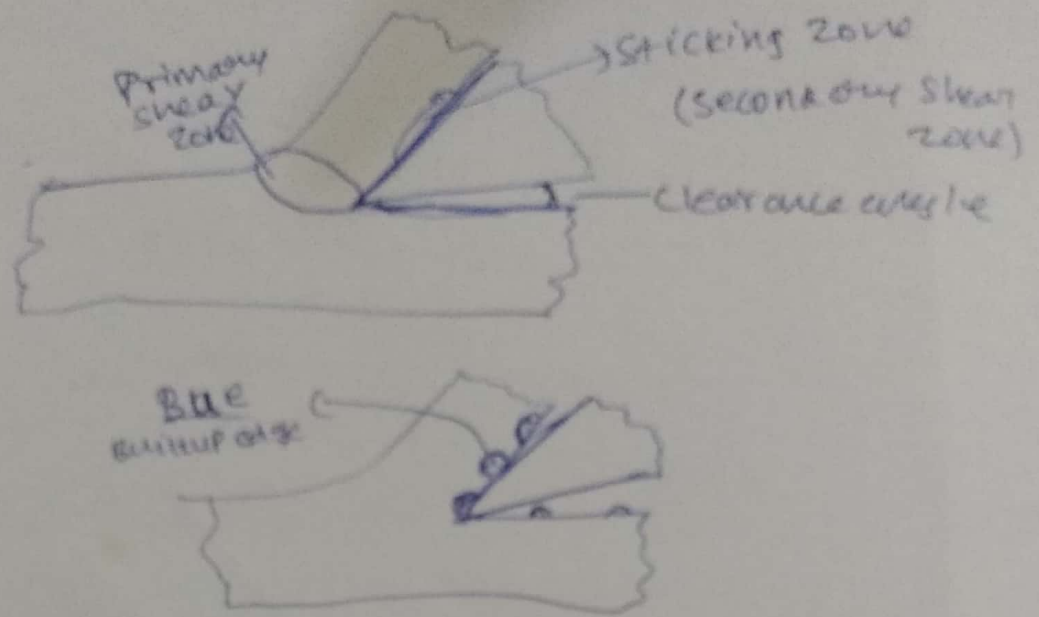
$$\tan \lambda = 0.5$$

$$\lambda = \tan^{-1}(0.5)$$

$$\lambda = 26.56$$

$$\alpha = 10^\circ$$

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When the zone under the cutting action is carefully examined, the following observations can be made

(i) The work layer deforms into a chip after it goes through a severe plastic deformation in the primary shear zone ~~just~~ ~~after its formation~~

Just after its formation, the chip flows over the rake surface of the tool and too strong adhesion b/w the tool and the heavily deformed chip surface results in ~~too~~ sticking of the chip material at this surface ~~under~~ ~~the~~ further plastic def. since despite no sticking, it flows. This zone is referred to as Secondary shear zone. Under suitable

condition, the machining operation is smooth and stable and produces continuous ribbon like chips.

At a somewhat high speed, the temp increases and the tendency of the plastically deformed material to adhere to the rake face increases and a lump is formed at the cutting edge. This is called a built up edge. It goes to a certain size but ultimately breaks due to the increased force exerted on it by the adjacent flowing material. After it breaks, the broken fragments adhere to the finished surface and the chips are first results in a rough finish.

These conditions for various types of chips.

(i) continuous chips without built up edge:

- Ductile Material
- small uncut thickness
- High cutting speed
- Large rake angle
- suitable cutting fluid.

(ii) Continuous chips with BUE:

- Stronger adhesion b/w chips and tool face.
- low rake angle.
- large uncut thickness.

(iii) Discontinuous chips:

- low cutting speed
- Brittle work material
- small rake angle
- large uncut thickness.

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Tuesday
Ass. Test

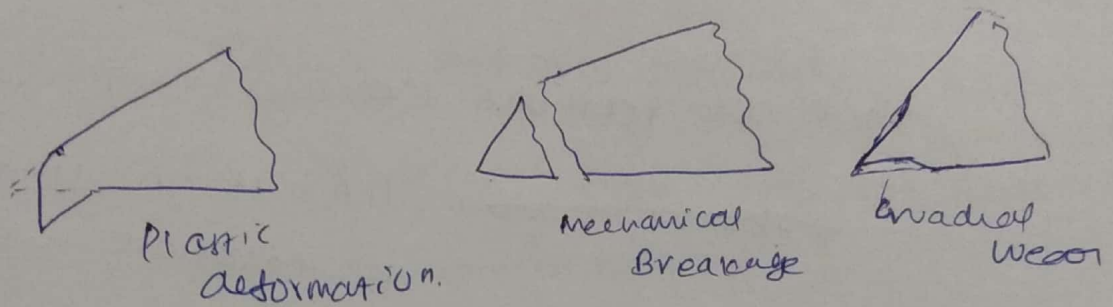
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Tool wear



The success of a machining process depends on the sharpness of the tool. When a cutting tool is unable to cut, consuming reasonable energy ~~and~~ and cannot produce an acceptable finish, it is considered to have failed. The failure of a cutting tool may be due to one or a combination of the following modes.

1. Plastic deformation of the tool due to high temperature and large stress.
2. Mechanical breakage of the tool due to large force and insufficient strength and toughness.
3. Blunting of the cutting edge of the tool to a process of gradual wear.

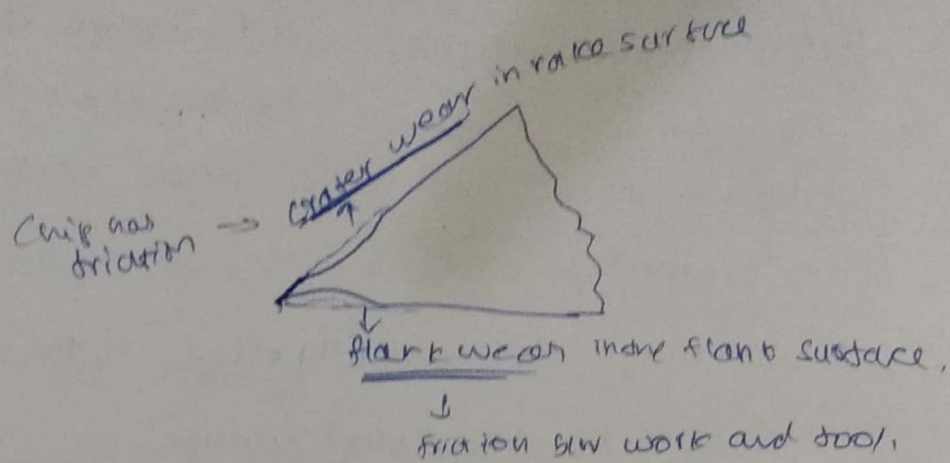


By a proper selection of the cutting tool material, tool geometry and cutting conditions, plastic deformation and mechanical failure can be prevented.

However the gradual wearing process cannot be totally stopped and ultimately the tool failure through wearing cannot be avoided.

The wearing action takes place on those surfaces along which there is a relative sliding with other surfaces. Thus the wear takes place on the rake surface where the chip flows over the tool and on the flank surfaces where

rubbing between the work & the tool occurs.



There are various mechanisms of wear.

- ~~1) Abrasion wear~~ 1) Abrasion wear
2) Adhesion wear
3) Diffusion wear.

Though all these mechanisms are normally active during a wear process, Abrasion and Adhesion are primarily responsible for flank wear.

Diffusion play an important role in the crater wear at high cutting speeds as temp rise is high in rake surface.

Cutting tool Materials

Since the machining operation is basically deformation process of the work material through the application of a force by the cutting

tool, The stability of the geometric form (form stability) is a key factor.

Obviously the cutting tool must (6 tool angles + nose radius).
Provide the maximum resistance to any tendency of alteration of its geometric shape. To achieve this, the cutting tool material must be properly selected.

The basic point is that the tool ^{material} must be harder than the work material.

normally 35% to 50% harder.

The Properties of ideal tool material can be summarised as follows.

(i) As far as possible, It should maintain its hardness which is sufficiently higher than that of the work at elevated temperature.

(ii) It should be tough enough to withstand shocks.

(iii) It should provide large resistance to the wearing action so excessive wear does not occur.

(iv) M ~~fric~~ b/w the work & tool should be low.

(v) Thermal conductivity & specific heat is high.

The materials commonly used for making cutting tools are

- (i) High Carbon Steel
- (ii) High speed steel
- (iii) Cemented carbide
- (iv) ceramics.

material	C	W	Cr	V	Mn	Fe
Carbon Tool steel	0.9				0.6	Rest (98.5%)
HSS	0.75	18	4	1.0	0.6	Rest (75.65%)

Cemented carbide - Carbides ~~are~~ of various

materials are bonded together using a cobalt Matrix

WC
Co
TaC
TiC

Tool Life and Machinability:

The major criteria for judging machinability is as follows

- (i) Machining forces and Power consumption.
- (ii) Surface finish
- (iii) Tool life.

Taylor's tool life equation,

Simplified eqn $\rightarrow VT^n = C$ n, C - constants

V - velocity

T - time tool life.

Generalised eqn

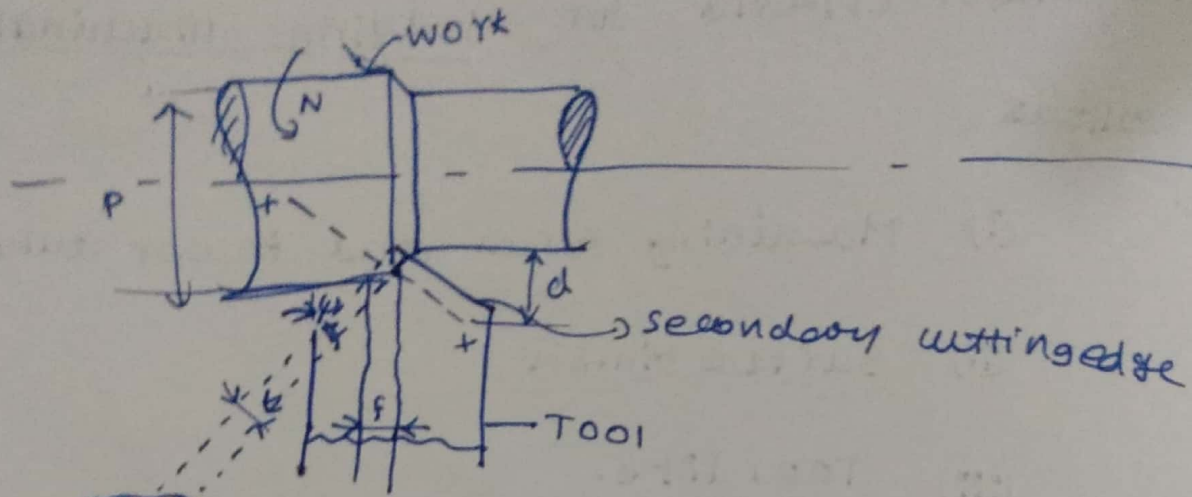
$$V = \frac{C'}{T^n t^p w^q}$$

$$VT^n t^p w^q = C'$$

t - uncut thickness

w - width of cut.

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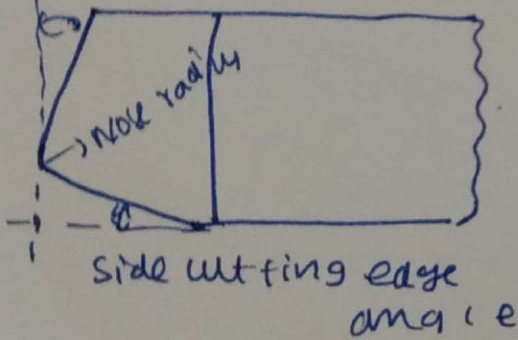
TOP view of

$$t_1 = f \cos \phi$$

$$w = \frac{d}{\cos \phi}$$

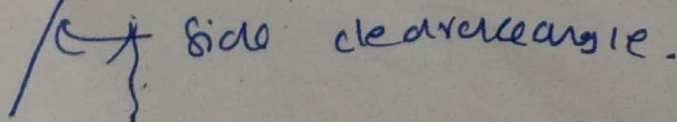
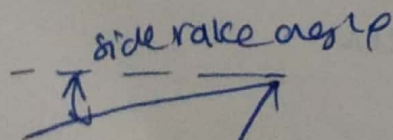
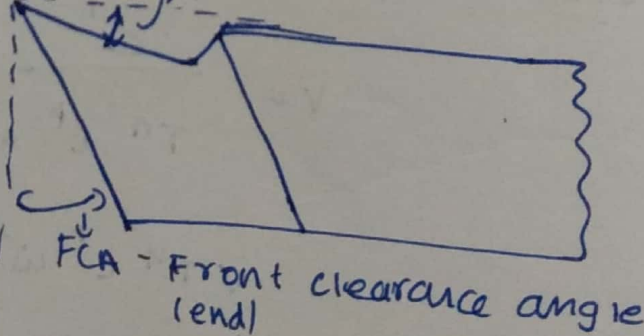
$$v = \frac{\pi D N}{1000} \text{ M/min}$$

FCE (Front cutting edge angle)

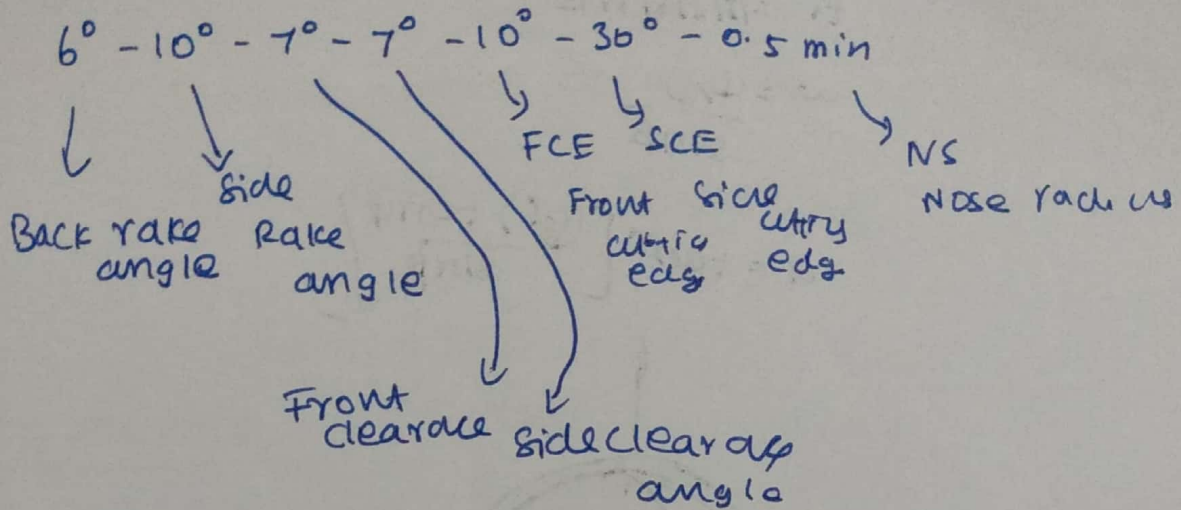


side view of

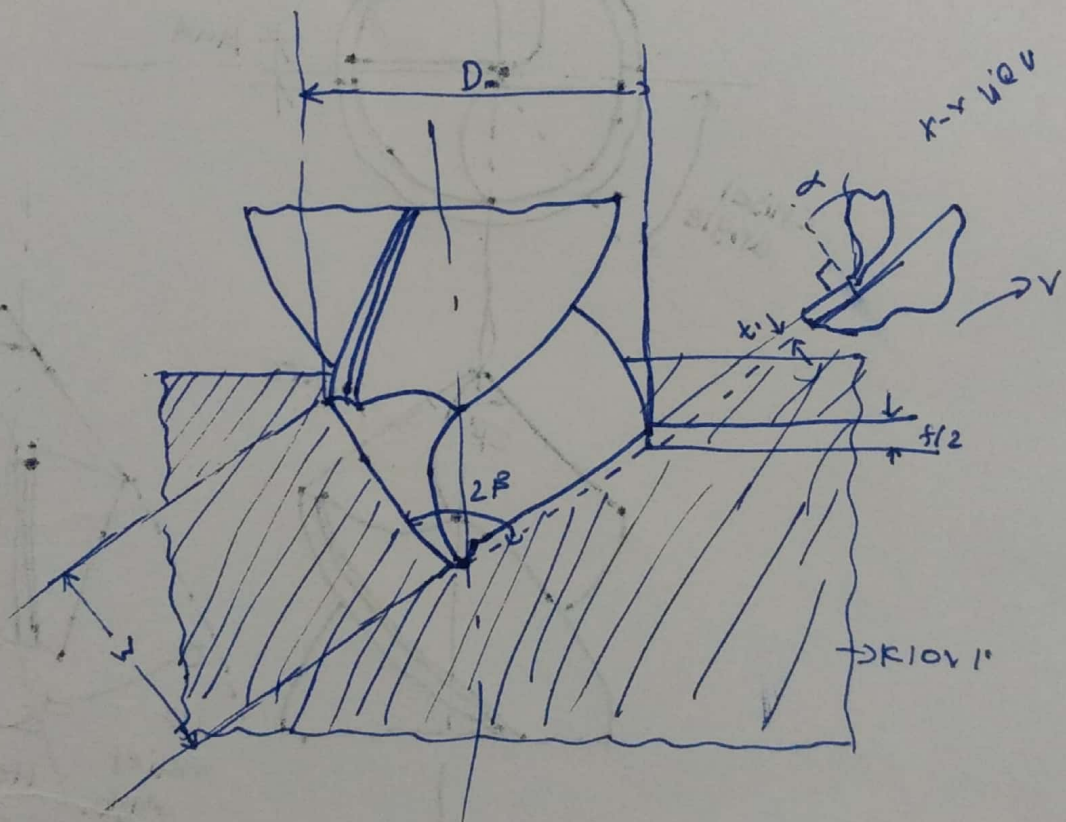
BRA (Back rake angle)



ASS (American Standard System) Tool signature



24110



The most common hole making operation is drilling. It is usually performed with drill bit, unlike shaping and turning, this involves 2 principal cutting edges. If the total advancement of the drill per revolution is f . Then stroke of each cutting edge is $f/2$ because each lift is setting the uncut air the

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glazing for a winding wheel.

Thermal aspects of cutting.

economics of cutting.

Cutting Fluids